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Evaluation of the reliability of non-coherent systems using Binary Decision Diagrams

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Abstract: While many systems are coherent, some systems have non-coherent structure due to their negative feedback loops and redundant loops. A system is non-coherent for a component if it is in a failed state when the component is in a working state, and when the component fails, the system is restored to the non-failed state. In this paper, we propose a straightforward method to evaluate the imprecise reliability of non-coherent systems using binary decision diagrams (BDD) and imprecise probabilities. The system components are categorized into three types according to their coherency inside the studied system: positively coherent, negatively coherent and non-coherent. The reliability of components is represented by intervals. The imprecise system reliability is obtained using operations performed on the different paths of the BDD. A numerical example is also given in order to illustrate the proposed method.

Keywords: Non-coherent Systems, Binary Decision Diagrams (BDD), Imprecise Reliability.

1. INTRODUCTION

The dependability of a system is defined as the property that enables the system users to place a justified confidence in the service it delivers. Reliability is one of the most important dependability attributes. It is defined as the ability of a system to function over an interval of time. However, the reliability analysis of non-coherent systems is limited and few real non-coherent systems have been studied in the literature. The non-coherent systems introduce difficulties in both qualitative and quantitative assessment of reliability compared with the coherent systems. If the system is non-coherent for component i, the system is in a failed state when component i is in a working state, and when component i fails the system is restored to the non-failed state. Therefore, system failure might occur after having repaired a failed component, or for a failed system the failure of an additional component may give a successful outcome of system performance. Non-coherent systems widely appears in fault trees with NOT gate such as multi-task systems (Andrews (2000)) and safety control applications.

Traditionally, reliability analysis of systems is tackled by using fault trees or reliability block diagrams based on minimal cut-sets or path-sets. In the last decades, Binary Decision Diagram (BDD) have provided an efficient method to represent and manipulate Boolean functions (Bryant (1986)), and are becoming an accurate and effective method in system reliability analysis (Rauzy (1993)) without the need of enumerating minimal cut-sets or path-sets of systems. A BDD represents the system structure function in a directed acyclic graph so that exact system reliability calculation can be conducted effectively using BDD.

Reliability data are an essential part of a reliability assessment. Indeed, system reliability is computed from the reliability of system components. Moreover, while computing system reliability, we need to consider not only components reliability but also the associated uncertainties, which can be divided into two types: aleatory uncertainty and epistemic uncertainty. Aleatory uncertainty (also called stochastic uncertainty) is caused by irreducible variability and the randomness of a stochastic phenomenon; epistemic uncertainty is due to the lack of information. Although it is reducible in theory, in real life epistemic uncertainty cannot be ignored (Aven (2011)), especially in cases with rare events (e.g. failures in nuclear reactors, failures in highly reliable systems such as railway systems).

To the best of our knowledge, there are very few works considering the use of imprecise probabilities in BDD. In Lehmann (2007), the authors proposed the use of belief functions theory in BDD by focusing on the definition of belief focal sets in the framework of BDD. This method was not applied in reliability assessment of systems. In Jacob et al. (2011), the authors proposed an approach based on interval computation methods in BDD-fault tree analysis, relying on the analysis of the structure and monotonicity of the Boolean formula. However, this method needs to check the monotonicity of the variables in the structure function that can make it inapplicable to very large systems. In 2015, a preliminary BDD-based method was proposed in Imakhlaf and Sallak (2016) to assess reliability of both coherent and non-coherent systems. However, this method was based on combination and projection operations, which are time consuming. In this paper, we will propose an original straightforward method to evaluate the imprecise system reliability based

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on BDD and basic probability assignments defined in the framework of belief functions theory without the need of using combination and projection operations. Moreover, this work focuses only on the assessment of reliability of non-coherent systems. Note that the proposed method is also adapted to coherent systems.

The rest of the paper is organized as follows. Section 2 introduces some basic definitions and properties of BDD. The basic definitions of belief functions theory, and a method to assess precise reliability of coherent system are respectively presented in Section 3. The definitions of non-coherent system, and our proposed method to evaluate the imprecise system reliability are detailed in section 4. Section 5 proposes a case study in order to show the effectiveness of the proposed approach. Finally, Section 6 concludes the work.

2. BINARY DECISION DIAGRAM

In this section we will give necessary background for BDD. The BDD construction, ordering and reduction are also described using some small examples.

2.1 Definitions

At first, we recall some basic graph-theoretical concepts.

Definition 1. A graph is a set of objects where some pairs of objects are connected by links, the links are called edges and the objects are called nodes.

Definition 2. A rooted graph is a graph in which one node has been distinguished as the root.

Definition 3. A directed graph is a graph in which the nodes are connected by edges associated with their directions.

Definition 4. A directed acyclic graph is a directed graph, in which there is no way to start at some node x and loops back to this node following a sequence of edges.

Definition 5. A Binary Decision Diagram (BDD) is a rooted directed acyclic graph such that:

(1) There are two types of terminal nodes: 0 and 1.
(2) Each internal node is labeled with a variable x_i and has two outgoing edges: $0-edge$ (dashed line) and $1-edge$ (solid line).

A BDD representation of a Boolean function f is based on Shannon's decomposition theorem:

Theorem 6. Let f be a Boolean function $\{0,1\}^n \to \{0,1\}$ depending on the set of variables $\{x_1, \dots, x_n\}$ then

$$f(x_1,\dots,x_n) = x_i.f(x_1,\dots,x_{i-1},1,x_{i+1},\dots,x_n) + \overline{x_i}.f(x_1,\dots,x_{i-1},0,x_{i+1},\dots,x_n) \quad (1)$$

- $f(x_1,\dots,x_{i-1},1,x_{i+1},\dots,x_n)$ is called the positive Shannon cofactor.
- $f(x_1,\dots,x_{i-1},0,x_{i+1},\dots,x_n)$ is called the negative Shannon cofactor.

It means that each Boolean function f can be decomposed into two sub-functions considering the cases where either x_i is equal to 0 or 1.

2.2 Construction

In this section, we show how to construct a BDD from a Boolean expression by a simple recursive procedure. More specifically, let us consider a Boolean function $f : \{0,1\}^n \to \{0,1\}$ depending on the set of variables $\{x_1, \dots, x_n\}$. We construct a BDD associated to f by applying Shannon's decomposition as follows:

First we chose a variable to be the root of the graph, so the first node is labelled x_i. Then, the $0-edge$ of x_i (dashed arrow) points to the sub-graph rooted by another variable x_j that represents $f(x_1,\dots,x_{i-1},0,x_{i+1},\dots,x_n)$. Next, we pursue with x_j until obtaining the terminal nodes.

On the other side, the $1-edge$ of x_i (solid arrow) points to the sub-graph rooted by another variable x_k that represents $f(x_1,\dots,x_{i-1},1,x_{i+1},\dots,x_n)$. Next, we pursue with x_k until obtaining the terminal nodes.

Example 7. In Fig. 1, we present the BDD construction of the Boolean function $f(x_1,x_2,x_3) = \overline{x_2} + x_1 \cdot x_3$ following the order $x_1 < x_2 < x_3$. This BDD can be also viewed as a compact representation of the Boolean function.

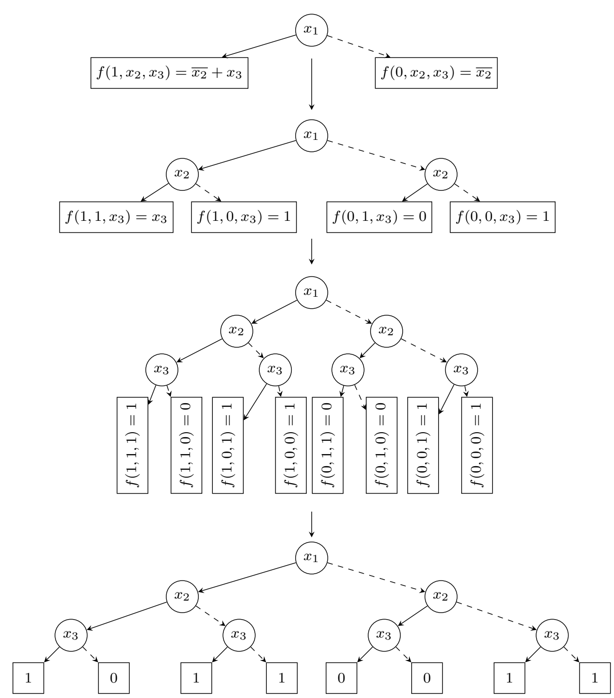

Fig. 1. Steps of BDD construction of $f(x_1,x_2,x_3) = \overline{x_2} + x_1 \cdot x_3$

2.3 Ordered Binary Decision Diagram

A major issue when constructing a BDD is to find an optimal variable ordering to avoid the exponential explosion of the number of nodes and also to speed up the algorithm that manipulate the BDD. The ordering of a BDD consists in associating to each level of the BDD graph a unique variable.

Definition 8. A Binary Decision Diagram is said to be ordered (OBDD) following the order $x_1 < \dots < x_n$, if on all paths through the graph of the BDD, the variables appear with respect to this order.

In other words, for any internal node $a = \Gamma(x_i, a_0, a_1)$ either a_0 is a terminal node or it is labelled with a variable x_j such that $x_i < x_j$, and either a_1 is a terminal node or it is labelled with a variable x_j such that $x_i < x_j$.

2.4 Reduced Ordered Binary Decision Diagram

Starting with a BDD satisfying the ordering property, we can reduce its size by repeatedly applying two transformation rules. We use the term Reduced Ordered Binary Decision Diagram (ROBDD) to refer to a maximally reduced BDD that follows some ordering. In the following, we give some formal definitions.

Definition 9. Two sub-graphs of a OBDD, F and G, are isomorphic if there exists a one-to-one function σ from the nodes of F onto the nodes of G such that for any node a if $\sigma(a) = a'$, then either both are non-terminal nodes with $index(a) = index(a')$, $\sigma(high(a)) = high(a')$, and $\sigma(low(a)) = low(a')$, or both are the same terminal nodes.

Definition 10. An OBDD is said to be reduced if $a_0 \neq a_1$ for any node $a = \Gamma(x_i, a_0, a_1)$, and there is no isomorphic sub-graphs.

More specifically, to reduce an OBDD we apply, repeatedly, the two following reduction rules as much as possible.

(1) All isomorphic sub-graphs are merged. In Fig. 2, x_1 points to two equivalent sub-graphs so at least one is useless.

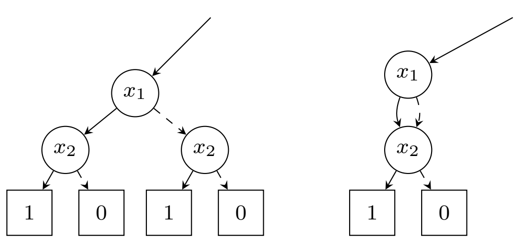

Fig. 2. First reduction rule

(2) For any node $a = \Gamma(x_i, a_0, a_1)$ if $a_0 = a_1$, we should remove the node a. As we can see in Fig. 3, the two edges of the node labelled x_1 point to the same node x_2, so the node x_1 is removed.

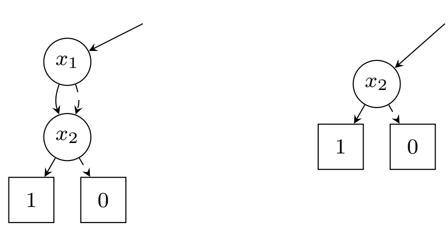

Fig. 3. Second reduction rule

The result of reduction depends only on the BDD to which it is applied and on the order of the variables. It is independent of the order of application of the rules. In practice, variable orderings are computed by heuristic algorithms and can sometimes be improved with local optimization and even simulated annealing algorithms.

3. PRECISE RELIABILITY OF SYSTEMS

In this section, we recall basic background of belief functions theory, and we explain how to use the BDD and belief functions theory in order to compute the precise system reliability.

3.1 Belief Functions Theory

In belief functions theory, the frame of discernment is the domain of definition and represents all possible sets of the variable of interest.

Definition 11. If Θ is a frame of discernment, then a function $m : 2^\Theta \to [0, 1]$ is called a basic probability assignment if:

$$\sum_{A \subset \Theta} m(A) = 1$$

The mass function $m(A)$ can be seen as the degree of belief assigned exactly to the realization of A. In probability theory, the masses can be assigned only to the singletons of Θ, whereas in belief functions theory, the masses can be assigned to the singletons and subsets of Θ.

In addition to masses, belief functions theory also has two important functions to represent knowledge: belief (bel) and plausibility (pl) functions.

The belief function $bel(A)$ is defined by:

$$bel(A) = \sum_{B \subseteq A} m(B)$$

The plausibility function $pl(A)$ is defined by:

$$pl(A) = \sum_{B \cap A \neq \emptyset} m(B)$$

In other words, $bel(A)$ is the part of belief that supports only A or its subsets, and $pl(A)$ is the part of belief that measures to what extent one fails to doubt A.

Therefore, belief and plausibility functions can be regarded as lower and upper bounds for the probability of A, i.e. $bel(A) \leq Pr(A) \leq pl(A)$.

For simplicity, let us consider a component i with two states. The frame of discernment Θ_i of the component i is then given by $\Theta_i = \{0_i, 1_i\}$, where 0_i and 1_i denote respectively the failed and operational states of the component i. The expert will provide the reliability data of the component i in the following form: $m_i(1_i)$ represents the degree of belief that the component i is in working state; $m_i(0_i)$ represents the degree of belief that the component i is in failure state; $m_i(\Theta_i) = 1 - m_i(0_i) - m_i(1_i)$ represents the epistemic uncertainty about the state of the component i. So the reliability r_i of a component i is expressed by :

$$\begin{aligned}[bel(1_i), pl(1_i)] &= [m_i(1_i), m_i(1_i) + m_i(\Omega_i)] \\ &= [m_i(1_i), 1 - m_i(0_i)]\end{aligned}$$

And the system reliability R_s is expressed by

$$\begin{aligned}[bel(1_s), pl(1_s)] &= [m_s(1_s), m_s(1_s) + m_s(\Omega_s)] \\ &= [m_s(1_s), 1 - m_s(0_s)]\end{aligned}$$

3.2 Reliability of coherent systems using BDD

In this section, we will compute the precise reliability of system in the absence of epistemic uncertainties, i.e., $m_i(\Omega_i) = 0$ for every i, hence $m_i(1_i) = 1 - m_i(0_i)$.

Let $\mathscr{A}$ be the set of paths from the root to 1, and $\mathscr{A}_p$ the set of variables present in a path p. We define $\mathscr{A}_p^+$ (resp. $\mathscr{A}_p^-$) as the set of variables present positively (resp. negatively) in the path p. Each binary variable x_i represents the state of the component i. The system reliability is given by (Jacob et al. (2011))

$$m(1_s) = \sum_{p \in \mathscr{A}} \left[\prod_{x_i \in \mathscr{A}_p^+} m_i(1_i) \prod_{x_i \in \mathscr{A}_p^-} m_i(0_i) \right] \qquad (2)$$

Example 12. For illustration, let us consider the parallel system $S1$ depicted in Fig. 4-a composed of two components 1 and 2. Its ROBDD is given in Fig. 4-b.

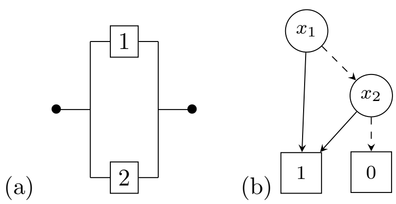

Fig. 4. (a) Reliability bloc diagram of $S1$ (b) ROBDD of $S1$

There are two paths from the root to 1. In the first path, x_1 is present positively and x_2 is not present. In the second path, x_1 is present negatively and x_2 is present positively. According to Eq. 2, the precise reliability of $S1$ is obtained easily by $R_{S_1} = m(1_S) = m_1(1_1) + m_1(0_1)m_2(1_2)$.

4. THE PROPOSED IMPRECISE BDD METHOD

In this section, we give some basic background of non-coherent systems. We also present an original BDD method to assess the reliability of non-coherent systems. This method is based on the classification of components into three categories. Finally, we discuss the importance of BDD variable ordering on the obtained system reliability results.

4.1 Basic background of non-Coherent systems

We recall that the state of a system S is related to the states of its components, a component i has only two states: a functioning state 1_i and a failed state 0_i. It is represented by a binary variables x_i.
We can determine the state of the system according to the states of its components by a structure function such that:

$$f(x_1, \ldots, x_n) = \begin{cases} 1 & \textit{if the system is working} \\ 0 & \textit{if the system is in a failed state} \end{cases}$$

A system is monotone if its structure function f is increasing, i.e.,$f(1_i, x) \geq f(0_i, x)$ where $f(1_i, x) = f(x_1, ..., x_{i-1}, 1, x_{i+1}, ..., x_n)$ and
$f(0_i, x) = f(x_1, ..., x_{i-1}, 0, x_{i+1}, ..., x_n)$.
A component i $(i = 1, ..., n)$ is relevant to the system state if $f(1_i, x) \neq f(0_i, x)$ for some x.
A system is coherent if it is monotone and all of its components are relevant.
In other words, a system is coherent if the improvement of component states cannot damage the system, and none of its components is irrelevant.
A **non-coherent system** is a system that does not satisfy the condition (1) or the condition (2), in other words, if it can move from a failed state to a working state by the failure of one of its components, or move from a working state to a failed state by repairing of one of its components. Non-coherent systems are becoming more common in multi-tasking systems, and safety control applications.
The elements in a non-coherent system can be classified into three types: single positive, single negated and double form (Contini et al. (2006)). In this work, we adopt another appellation: positively coherent, negatively coherent, and non coherent.
Types of elements are identified according to the sign of the form in which the elements appear and can be easily obtained from the structure of the structure function.
Formally, a component i is called **positively coherent** if

$$f(x_1, ...x_{i-1}, 1, x_{i+1}..., x_n) - f(x_1, ...x_{i-1}, 0, x_{i+1}..., x_n) \geq 0$$

for all $x_1, ...x_{i-1}, x_{i+1}..., x_n \in \{0, 1\}$. Hence, the system reliability is an increasing function of the component reliability $r_i \in [0, 1]$ which is equivalent to the following condition:
When a positively coherent element is repaired $(0_i \to 1_i)$, the system's reliability is improved; when it fails $(1_i \to 0_i)$, the system reliability declines. In coherent systems, all elements are positively coherent.
A component i is called **negatively coherent** if the following condition is satisfied

$$f(x_1, ...x_{i-1}, 1, x_{i+1}..., x_n) - f(x_1, ...x_{i-1}, 0, x_{i+1}..., x_n) \leq 0$$

for all $x_1, ...x_{i-1}, x_{i+1}..., x_n \in \{0, 1\}$. Then the system reliability is a decreasing function of the component reliability $r_i \in [0, 1]$. When a negatively coherent element is repaired $(0_i \to 1_i)$, the system's reliability declines. When it fails $(1_i \to 0_i)$, the system reliability is improved.
Otherwise, the component is called **non-coherent** for the system. In this case the impact of repair or failure of this element depends on the states of other elements. It means that if a non-coherent component move from its state in time t to another state in $t+1$, we cannot predict the state of the system in $t + 1$. We summarize these definitions in Table 1.

4.2 Estimation of reliability of non-coherent systems

Let $\mathcal{MP}$ and $\mathcal{MC}$ be the set of paths from the root to 1 and 0 respectively, and $\mathcal{P}$ and $\mathcal{C}$ the set of variables present in a path $\mathcal{MP}_i$ and $\mathcal{MC}_i$. In the following, we consider $\mathcal{P}_i^+$, $\mathcal{P}_i^-$, and $\mathcal{P}_i^{+-}$ as the set of variables representing respectively positively coherent, negatively coherent, and non-coherent components in each path $\mathcal{MP}_i$. And $\mathcal{C}_i^+$, $\mathcal{C}_i^-$, and $\mathcal{C}_i^{+\setminus-}$ as the set of variables representing respectively positively coherent, negatively coherent, and non-coherent components in each path $\mathcal{MC}_i$. The system reliability can be formulated as $R_s = [m(1_s), 1 - m(0_s)]$ where:

$$m(1_s) = \sum_{\mathcal{MP}_j \in \mathcal{MP}} [\prod_{x_i \in \mathcal{P}_j^+} x_i.m_i(1_i) + (1-x_i).(1-m_i(1_i)). \quad (3)$$
$$\prod_{x_i \in \mathcal{P}_j^-} x_i.(1-m_i(0_i)) + (1-x_i).m_i(0_i)).$$
$$\prod_{x_i \in \mathcal{P}_j^{+\setminus-}} [x_i.min(m_i(1_i), 1-m_i(0_i))+$$
$$(1-x_i).min(1-m_i(1_i), m_i(0_i))]]$$

$$m(0_s) = \sum_{\mathcal{MC}_j \in \mathcal{MC}} [\prod_{x_i \in \mathcal{C}_j^+} x_i.(1-m_i(0_i)) + (1-x_i).m_i(0_i). \quad (4)$$
$$\prod_{x_i \in \mathcal{C}_j^-} x_i.m_i(1_i) + (1-x_i).(1-m_i(1_i))).$$
$$\prod_{x_i \in \mathcal{C}_j^{+\setminus-}} [x_i.min(m_i(1_i), 1-m_i(0_i))+$$
$$(1-x_i).min(1-m_i(1_i), m_i(0_i))]]$$

Note that a non-coherent component i could be coherent positively or negatively according to the state of the rest of components. So, we take the minimum of the two masses which can be affected to the component i if it was positively or negatively coherent. We can simplify the equations (3) and (4), because we have $1 - m(0_i) = m(1_i) + m(\Omega_i)$, and $1 - m(1_i) = m(0_i) + m(\Omega_i)$ hence $min(m_i(1_i), 1 - m_i(0_i)) = m_i(1_i)$, and $min(m_i(0_i), 1 - m_i(1_i)) = m_i(0_i)$. So, the system reliability is finally expressed by:

Table 1. Definition of components type

Component			System	
Coherence	state at t	$t+1$	state at t	$t+1$
+	0	1	1	1
			0	$\{0,1\}$
	1	0	1	$\{0,1\}$
			0	0
-	0	1	1	$\{0,1\}$
			0	0
	1	0	1	1
			0	$\{0,1\}$
+/-	0	1	1	$\{0,1\}$
			0	$\{0,1\}$
	1	0	1	$\{0,1\}$
			0	$\{0,1\}$

$$m(1_s) = \sum_{\mathcal{MP}_j \in \mathcal{MP}} [\prod_{x_i \in \mathcal{P}_j^+} x_i.m_i(1_i) + (1-x_i).(1-m_i(1_i)). \quad (5)$$
$$\prod_{x_i \in \mathcal{P}_j^-} x_i.(1-m_i(0_i)) + (1-x_i).m_i(0_i)).$$
$$\prod_{x_i \in \mathcal{P}_j^{+\setminus-}} x_i.m_i(1_i) + (1-x_i).m_i(0_i)]$$

$$m(0_s) = \sum_{\mathcal{MC}_j \in \mathcal{MC}} [\prod_{x_i \in \mathcal{C}_j^+} x_i.(1-m_i(0_i)) + (1-x_i).m_i(0_i). \quad (6)$$
$$\prod_{x_i \in \mathcal{C}_j^-} x_i.m_i(1_i) + (1-x_i).(1-m_i(1_i))).$$
$$\prod_{x_i \in \mathcal{C}_j^{+\setminus-}} x_i.m_i(1_i) + (1-x_i).m_i(0_i)]$$

In the absence of epistemic uncertainties, i.e., $m_i(1_i) = 1 - m_i(0_i)$ for every i, we find the precise system reliability given by the precise BDD method (cf. Eq. 2):

$$m(1_s) = \sum_{\mathcal{MP}_j \in \mathcal{MP}} [\prod_{x_i \in \mathcal{P}_j} x_i.m_i(1_i) + (1-x_i).(1-m_i(1_i))]$$

Where $\mathcal{P}_j$ is the set of variables in the path $\mathcal{MP}_j$.
Note that, in the absence of epistemic uncertainties, we obtain the same precise system reliability for all the variable ordering. However, in the imprecise case, we could obtain different intervals depending on the choice of variable ordering. To be cautious (pessimistic view), we should choose the more conservative one.

5. CASE STUDY

Let us consider a pumping system S that consists of a tank, two pumps P_1 and P_2, and a valve V. The system works when the water is pumped from the tank at a rate between $1l/s$ and $3l/s$. The pump P_1 fails by pumping too hard, and the pump P_2 fails by not pumping hard enough.

If the pump P_1 pumps too hard, the valve V has to switch from the pump P_1 to the pump P_2 and the system works. The valve fails if it does not switch in this case. If the pump P_1 is pumping too hard, the valve is failing (we cannot switch to P_2), and we have a leak, the system is still working (we conserve a rate between $1l/s$ and $3l/s$) (cf. Figure 5).

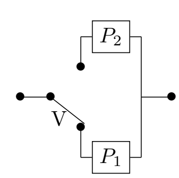

Fig. 5. The pumping system S

In the sequel, for simplicity, the working state of a component X is denoted by X and its failure is denoted by $\overline{X}$.

The component P_1 is negatively coherent, the component P_2 is positively coherent, and the component V is non-coherent. The Boolean function is given by:

$$\varphi(P_1, P_2, V) = V.P_2 + \overline{V}.\overline{P_1}$$

The reliabilities of components are given by:$r_{P_1} \in [0.9, 0.95]$, $r_{P_2} \in [0.9, 0.91]$, and $r_V \in [0.8, 0.95]$.
In the following, we will construct the BDDs of the system S with respect to all possible variable ordering and we will compute the reliability of the system using each BDD. Finally, we compare these results with the exact solution obtained with Jacob's method Jacob et al. (2011). we

construct the first BDD (c.f. Figure 6-a) with respect the order $V < P_1 < P_2$. In this case, there are 2 paths from the root to 1 ($0_V 0_{P_1}$ and $1_V 1_{P_2}$), and 2 from the root to 0 ($0_V 1_{P_1}$, and $1_V 0_{P_2}$). Then, according to equations (6) and (7) we obtain:

$$m_s(1_s) = m_V(0_V)m_{P_1}(0_{P_1}) + m_V(1_V)m_{P_2}(1_{P_2})$$
$$m_s(0_s) = m_V(0_V)m_{P_1}(1_{P_1}) + m_V(1_V)m_{P_2}(0_{P_2})$$

So the system reliability is given by the following interval:

$$r_s \in I = [0.7225, 0.883]$$

On the other hand, if we change the variable ordering from $V < P_1 < P_2$ to $P_1 < P_2 < V$, we obtain the BDD in Figure 6-b, and the results are:

$$\begin{aligned} m'_s(1_s) =& m_V(1_V)m_{P_2}(1_{P_2})\left[m_{P_1}(1_{P_1}) + m_{P_1}(\Omega_{P_1})\right] + \\ & m_{P_1}(0_{P_1})m_V(0_V)\left[m_{P_2}(0_{P_2}) + m_{P_2}(\Omega_{P_2})\right] + \\ & m_{P_1}(0_{P_1})m_{P_2}(1_{P_2}) \end{aligned}$$

$$\begin{aligned} m'_s(0_s) =& m_{P_1}(1_{P_1})m_V(0_V)\left[m_{P_2}(1_{P_2}) + m_{P_2}(\Omega_{P_2})\right] + \\ & m_{P_2}(0_{P_2})m_V(1_V)\left[m_{P_1}(0_{P_1}) + m_{P_1}(\Omega_{P_1})\right] + \\ & m_{P_1}(1_{P_1})m_{P_2}(0_{P_2}) \end{aligned}$$

According to this order, the system reliability is given by

$$r_s \in I' = [0.72925, 0.87085]$$

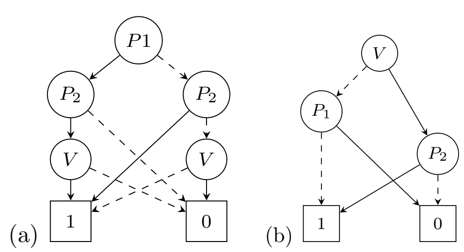

Fig. 6. (a) ROBDD of S with the order $P_1 < P_2 < V$ (b) ROBDD of S with the order $V < P_1 < P_2$

Finally, we assess the intervals of system reliability according to all possible orders. There are 3! different ways of ordering. Figure 7 summarizes the obtained results.

We obtain the same interval I' with both the orders $P_1 < P_2 < V$ and $P_2 < P_1 < V$. The interval I' is the less conservative one. On the other hand, we obtain the interval I with $P_1 < V < P_2$, $P_2 < V < P_1$, $V < P_1 < P_2$, and $V < P_2 < P_1$. The exact solution obtained by using all the upper and lower bounds of reliabilities of components is $r_s \in [0.73, 0.8695]$. As we can see (cf. Figure 7), the exact solution (red interval) is always included in all the intervals (blue intervals) obtained by our method which proves its correctness and efficiency in this case study. Moreover, the exact solution cannot be obtained for systems with large number of components because it is NP-hard problem, whereas our method provided formula that can be applied for all types of systems. To be cautious, we should choose the interval I, and thus the reliability of system is finally given by $r_s \in I = [0.7225, 0.883]$.

6. CONCLUSION

In this paper, we have proposed a method to assess the reliability of non-coherent system in the presence of epistemic uncertainty. We have distinguished components into three categories positively coherent components, negatively coherent components, and non-coherent components. The

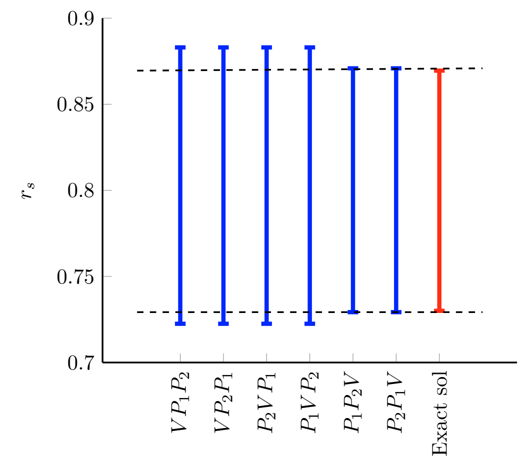

Fig. 7. Intervals of system reliability

method that we propose provides non-coherent system reliability expressed by an interval, which bounded the exact solution. However, as we have shown, the intervals depend on the variable ordering. In the future, we will assess conditions under which, the variable ordering will not influence the length of the obtained system reliability intervals.

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REFERENCES

Andrews, J.D. (2000). To not or not to not. *Proc. Int. System Safety Conf.*, 267275.

Aven, T. (2011). Interpretations of alternative uncertainty representations in a reliability and risk analysis context. *Reliability Engineering & System Safety*, 96(3), 353–360.

Bryant, R. (1986). Graph-based algorithms for boolean function manipulation. *IEEE Transactions on computers*, 100(8), 677–691.

Contini, S., Cojazzi, G., and de Cola, G. (2006). On the exact analysis of non-coherent fault trees: The ASTRA package (PSAM-0285). In *Proceedings of the Eighth International Conference on Probabilistic Safety Assessment & Management (PSAM)*, 1813–1821. ASME International.

Imakhlaf, A. and Sallak, M. (2016). An extended bdd method for the assessment of system reliability under aleatory and epistemic uncertainties. In *ESREL 2016*, 2751–2756.

Jacob, C., Dubois, D., and Cordoso, J. (2011). *Uncertainty Handling in Quantitative BDD-Based Fault-Tree Analysis by Interval Computation*, 205–218. Springer Berlin Heidelberg.

Lehmann, N. (2007). Shared ordered binary decision diagrams for dempster-shafer theory. In *European Conference on Symbolic and Quantitative Approaches to Reasoning and Uncertainty*, 320–331. Springer.

Rauzy, A. (1993). New algorithms for fault trees analysis. *Reliability Engineering & System Safety*, 40(3), 203–211.